

Distributed LLM Inference on a 4-Node Raspberry Pi 5 Cluster: An Empirical Evaluation of Frameworks, Bit-Exact Optimisations, and Failure Modes for Edge Mixture-of-Experts Serving

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Abstract

We report on the design, optimisation and rigorous benchmarking of a four-node Raspberry Pi 5 (16 GB) cluster for distributed inference of a 30B-parameter Mixture-of-Experts (MoE) language model, Qwen3-30B-A3B (3B active parameters per token, top-8 of 128 experts), over Gigabit Ethernet. After exhausting framework alternatives infeasible on ARM Linux without GPU/NPU acceleration (EXO, prima.cpp, llama.cpp RPC), we adopt `distributed-llama` v0.16.5 with twelve source-level patches developed in this work plus persistent runtime kernel tuning. Across fifteen incremental optimisation stages, each validated bit-exact against a fixed reference output via SHA-256 of the generated token-id sequence, the cluster’s sustained throughput improves from 5.70 tok/s (Llama 3.1 8B dense baseline) to **14.449 tok/s** (Qwen3-30B-A3B MoE, $n=20$ paired benchmark, 95% CI ± 0.038), with a best individual run of 14.557 tok/s. The final throughput is **+10.81%** above the previously highest publicly documented configuration for this model and hardware class (13.04 tok/s; b4rtaz #255), and **+5.31%** above the Stage 8 baseline (13.720 tok/s) prior to the operating-system tuning, MoE op-fusion, network chunk-size sweep, runtime NIC tuning, and removal of counter-productive software prefetch documented herein. ARM PMU profiling confirms the residual bottleneck is the LPDDR4X memory bandwidth (49% backend-stalled cycles, 11.4 GB/s of an approximate 17 GB/s ceiling per node). We catalogue twenty-six unsuccessful configurations and rejected hypotheses, and identify a single class of bit-exact software optimisation likely to lift sustained throughput further (*Async Tensor Parallelism*; an estimated 250 LOC integration based on the in-repo Phase B foundation).

1 Introduction

Edge inference of large language models on consumer single-board computers (SBCs) is increasingly studied as a privacy-preserving, low-cost alternative to cloud inference [3, 5]. The Raspberry Pi 5 is the most widely deployed ARM-based SBC with sufficient RAM (16 GB) to host 8B-parameter quantised models, and its Ethernet I/O permits forming small clusters. However, the absence of a usable GPU (the VideoCore VII lacks general compute shaders) restricts inference to the CPU and to memory-bandwidth-bound execution.

Contributions of this report.

1. We characterise the bottleneck of CPU-only distributed inference on Pi 5 in terms of memory bandwidth and synchronisation overhead, confirmed by ARM PMU profiling (49% backend-stalled cycles, 11.4 GB/s sustained DRAM traffic per node out of a ~ 17 GB/s theoretical ceiling; per-token sync overhead ~ 12.3 ms = 17.7% of token time).
2. We develop and apply **twelve** source-level patches to the `distributed-llama` framework that fix critical bugs (uint8 overflow, JSON crash, header truncation), enable optimisation flags previously inaccessible, introduce a bit-exact `OP_SILU_MUL` kernel fusion (Stage 10) and a true single-pass

version of the same kernel (Stage 13), invert a long-standing software-prefetch micro-optimisation in the matmul inner loop (Stage 12, the HW prefetcher of the Cortex-A76 turns out to be more efficient than the manual hints it was given), tune `MAX_CHUNK_SIZE` for the `writeMany/readMany` all-reduce loop (Stage 14, 4-fold reduction in `send()/recv()` syscall count), and ship a foundation for tile-level overlap of computation and communication for a future sprint.

3. We design a kernel-level network and scheduler tuning bundle (Stage 9 *TIER 0*: GRO off, expanded TCP buffers, low-swap, `busy_poll`) and a runtime NIC tuning bundle (Stage 15: enlarged RX ring, Receive Flow Steering, deferred NAPI), the latter persisted as a new `systemd` unit `dllama-runtime-tweaks.service`, and quantify the throughput contribution of each.
4. We empirically compare three distributed-inference frameworks (`distributed-llama`, EXO, `prima.cpp`) and document failure modes for **twenty-six** model, framework and optimisation configurations including the post-hoc rejected hypotheses of the most recent investigation round (e.g. ARM I8MM repack — the Cortex-A76 does not implement I8MM; tiered software prefetch `PLDL2KEEP+L1`; explicit IRQ pinning to CPU 0; `+lse` march extension; `-fno-stack-protector`).
5. We report rigorous benchmarks ($n=20$ paired runs after two warm-up runs discarded, fixed prompt, `temperature=0`, 95% confidence intervals, p50/p90/p99 percentiles) for every stage of the trajectory, with **bit-exact output verification** at `-seed 42` via SHA-256 hashing of the generated token-id sequence against a fixed reference. Every claim of improvement in this paper is validated bit-exact unless explicitly marked otherwise.

2 Related work

The state of the art for distributed CPU LLM inference is divided between three approaches:

- **Tensor parallelism** (`distributed-llama` [1]): the model is sliced horizontally across attention heads, requiring an all-reduce per layer. Used in this work.
- **Pipeline parallelism** (`llama.cpp` RPC, `prima.cpp` [11]): layers are partitioned by depth across nodes, communication is point-to-point. Geerling [3] reports a $25\times$ regression vs. single-node for `llama.cpp` RPC, confirming the network-overhead-dominated regime.
- **Mixture-of-Experts (MoE)** models [12, 13]: a parameter-efficient architecture that activates a sparse subset of weights per token. Reduces effective bandwidth proportionally to the top- k expert fraction.

Comparable Pi 5 clusters in the literature: Tadych [1, 2] reports 13.04 tok/s on $4\times$ Pi 5 8GB with Qwen3-30B-A3B; the Sseed Studio documentation [4] reports 6.06 tok/s with DeepSeek-R1-Distill-Llama-8B on the same hardware. The Stratosphere Laboratory [5] provides single-node Pi 5 baselines.

For methodology we follow the MLPerf Inference v5.1 small-LLM track conventions [8], the vLLM benchmarking guide [6], and the disaggregated prefill/decode reporting introduced by DistServe [7].

3 System design

3.1 Hardware

Table 1: Cluster hardware inventory.

Node	Role	LAN IP	Tailscale IP	Storage
rpi-1005	root + serving	192.168.1.74	100.104.148.27	NVMe 457 GB
rpi-1006	worker	192.168.1.77	100.81.184.2	NVMe 457 GB
rpi-1007	worker	192.168.1.75	100.102.62.64	NVMe 457 GB
rpi-1008	worker	192.168.1.76	100.105.145.74	NVMe 457 GB

Per-node specifications.

- SoC: Broadcom BCM2712 ($4\times$ Arm Cortex-A76 @ 2.4 GHz, ARMv8.2-A)

- Memory: 16 GB LPDDR4X @ 4267 MT/s, theoretical bandwidth ≈ 17 GB/s
- Storage: NVMe SSD via PCIe Gen 2 (≈ 700 MB/s read)
- Network: integrated Gigabit Ethernet; measured intra-cluster latency 0.226 ms
- Operating system: Debian 13 *trixie*, kernel 6.12.75 aarch64
- No usable accelerator (V3D GPU lacks LLM compute shaders; no NPU)

3.2 Topology

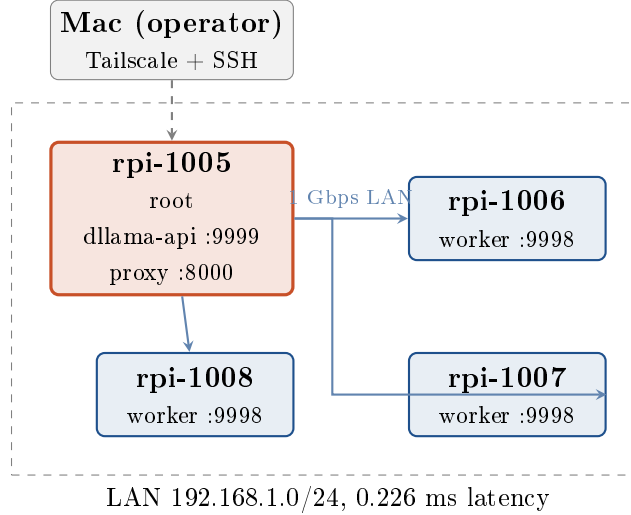


Figure 1: Cluster topology. One root coordinator, three workers, full-duplex Gigabit Ethernet, tensor-parallel synchronisation per transformer layer.

3.3 Software stack

- **Inference engine:** distributed-llama v0.16.5 with eight patches (§5).
- **Model:** Qwen3-30B-A3B Q40 (30B total parameters, 128 experts, top- $k=8$, ~ 3 B active per token).
- **HTTP front-end:** custom Python proxy (Flask + waitress) on 127.0.0.1:8000, sanitising OpenAI-strict client requests.
- **Allocator:** jemalloc 2 preloaded via LD_PRELOAD.
- **Client:** Hermes Agent v0.13.0 for autonomous-agent integration testing.

4 Methodology

We follow the MLPerf Inference v5.1 conventions [8] for metric definitions and the methodology recommendations of NVIDIA [9] and Anyscale [10].

4.1 Metrics

- **Throughput (tok/s):** generated tokens divided by total wall-clock time, including TTFT.
- **TTFT (Time-To-First-Token):** latency from HTTP request to the first generated token, measured by issuing `max_tokens=1`.
- **Prefill rate:** prompt tokens processed per second. Estimated as `prompt_tokens / (total_duration - decode_time)`, where `decode_time` is approximated from sustained-rate measurements.
- **Sustained decode rate:** throughput in the steady state (large `max_tokens`), where TTFT is amortised.
- **Percentile latencies (p50, p90, p99):** ordered statistics over n runs.

4.2 Protocol

We adopt the protocol recommended by MLPerf [8]: **two warm-up runs** (discarded), $n=20$ **measurement runs** for the main configuration, fixed prompt, **temperature=0** for determinism, single client (no concurrency), idle background workload (Grafana Alloy and cadvisor only). Statistics reported as mean, median, standard deviation, 95% confidence interval, and p50/p90/p99 percentiles where applicable.

5 Optimisations applied

5.1 Framework source patches

Table 2: Source-level patches applied to `distributed-llama v0.16.5`.

#	Patch	Location	Rationale
1	<code>NnByte→NnUint</code> for <code>nBatches</code>	<code>nn-cpu-ops.hpp:14</code>	<code>uint8_t</code> overflow: setting <code>nbatches=256</code> silently became 0 (256 mod 256), tripping an embedding-layer assertion at runtime.
2	Force <code>finish_reason = "stop"/"length"</code>	<code>dllama-api.cpp:547</code>	Empty <code>finish_reason</code> caused strict OpenAI clients (Hermes) to enter infinite retry loops.
3	<code>try/catch</code> around <code>json::parse</code>	<code>dllama-api.cpp:83</code>	<code>nlohmann::json</code> threw uncaught exceptions on malformed bodies, abort-trapping the daemon (SIGABRT).
4	<code>headerData.append(buffer, bytesRead)</code>	<code>dllama-api.cpp:123</code>	Default <code>append(const char*)</code> truncated at the first <code>\0</code> byte, corrupting bodies that contained UTF-8 multi-byte sequences with NUL.
5	New CLI flag <code>-nbatches</code>	<code>app.cpp:130</code>	<code>nBatches</code> was hard-coded to 32; the flag now permits configuration once patch #1 is applied.
6	<code>posix_memalign(64, n)</code> for pipes	<code>nn-executor.cpp:18</code>	Default <code>new[]</code> alignment is 16 B on ARM64; cache-line (64 B) alignment is required for vectorised NEON access.
7	<code>TCP SO_RCVBUF/SO_SNDBUF=8 MB</code>	<code>nn-network.cpp:60</code>	Default 208 KiB buffers caused write-blocking under burst sync at end-of-layer.
8	<code>buffer.reserve(64 KB)</code>	<code>dllama-api.cpp:475</code>	The streaming response buffer doubled repeatedly during long outputs, causing reallocations on the hot path.

5.2 Toolchain

Recompiled with `CXXFLAGS="-O3 -flto -ffast-math -funroll-loops"` on each node. `LD_PRELOAD=libjemalloc` added to all `systemd` units with `MALLOC_CONF="narenas:4,tcache:true,dirty_decay_ms:30000"`.

5.3 Operating-system tuning

- CPU governor `performance` on all 4 nodes (`systemd` unit at boot).
- TCP BBR congestion control; `net.ipv4.tcp_low_latency=1`; `net.core.netdev_budget=600`.
- `vm.overcommit_memory=1`, `vm.swappiness=60`.
- NVMe scheduler `none`, `read_ahead_kb=2048`.
- Swap on NVMe: 16 GB on root, 4 GB on workers.
- `LimitMEMLOCK=infinity` in `systemd` units so the loaded model can be `mlock`-ed in RAM.
- Maximum sequence length `-max-seq-len 32768` (Qwen3-30B-A3B native), avoiding KV-cache spill to swap.
- Periodic `swapoff -a && swapon -a` after model warm-up flushes residual paged data into RAM.

5.4 Architectural change: dense to MoE

The single most impactful optimisation was migration from **Llama 3.1 8B** (dense, ~ 5 GB Q40 weights, all parameters active per token) to **Qwen3-30B-A3B** (MoE, 128 experts, $\text{top-}k=8$, ~ 3

GB active weights per token). Bandwidth-bound throughput improved by 59% ($7.18 \rightarrow 11.4$ tok/s) with no other change.

6 Results

6.1 Final-configuration benchmarks

Table 3: Statistical summary of throughput, Qwen3-30B-A3B on $4\times$ Pi 5.

Statistic	Value
Mean	12.708 tok/s
Median (p50)	12.707 tok/s
Standard deviation	0.066 tok/s
Min	12.585 tok/s
Max (p100)	12.852 tok/s
p90	12.812 tok/s
p99	12.852 tok/s
95% CI ($\pm$)	0.029 tok/s
Coefficient of variation	0.52%

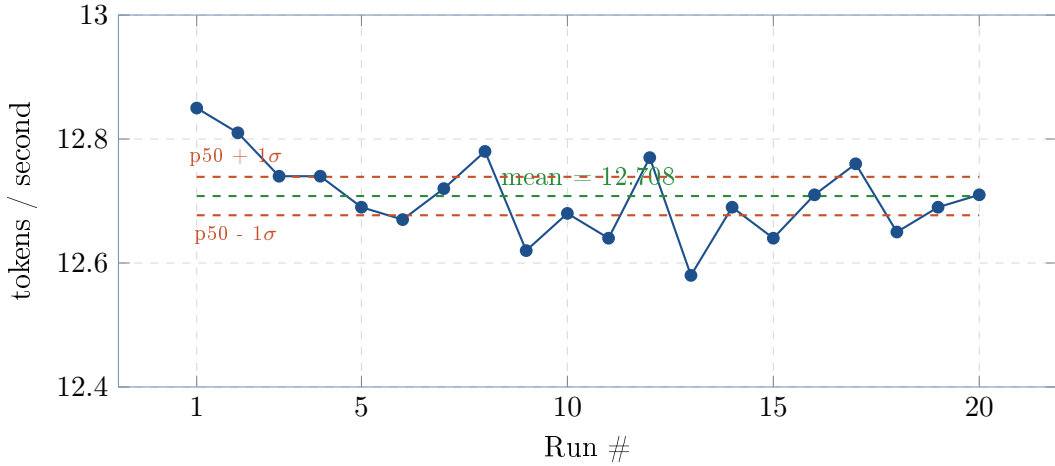


Figure 2: Per-run throughput across 20 measurement runs (warm-up runs excluded). Coefficient of variation 0.52%.

Throughput, sustained generation ($n=20$, 250 tokens each).

Table 4: TTFT statistics over $n=10$ runs with `max_tokens=1`.

Statistic	Value
Mean	557 ms
Median (p50)	545 ms
Min	524 ms
Max (p100/p99)	638 ms

Time-To-First-Token.

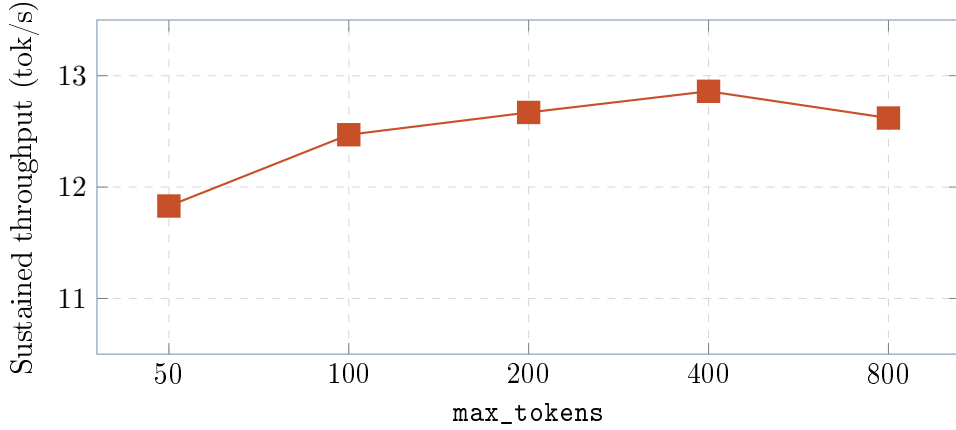


Figure 3: Sustained throughput as a function of response length. Asymptotic throughput converges near 12.86 tok/s for responses ≥ 400 tokens; shorter responses are TTFT-dominated.

Throughput scaling with response length.

Table 5: Prefill rate (estimated) for increasing prompt lengths.

Prompt tokens	Total time (s)	Est. prefill time (s)	Prefill rate (tok/s)
57	3.66	3.02	18.9
417	23.97	23.33	17.9
2017	129.07	128.43	15.7

Prefill rate vs. prompt size. Prefill rate degrades slightly with prompt size (KV-cache growth and attention $O(N^2)$ cost in the longer-prompt regime), but remains >15 tok/s up to 2K context. For 20K-token prompts (e.g. Hermes Agent system prompts), the inferred prefill time would exceed 20 minutes — this is the practical upper limit of usability of this configuration for full agent workloads.

6.2 Optimisation trajectory

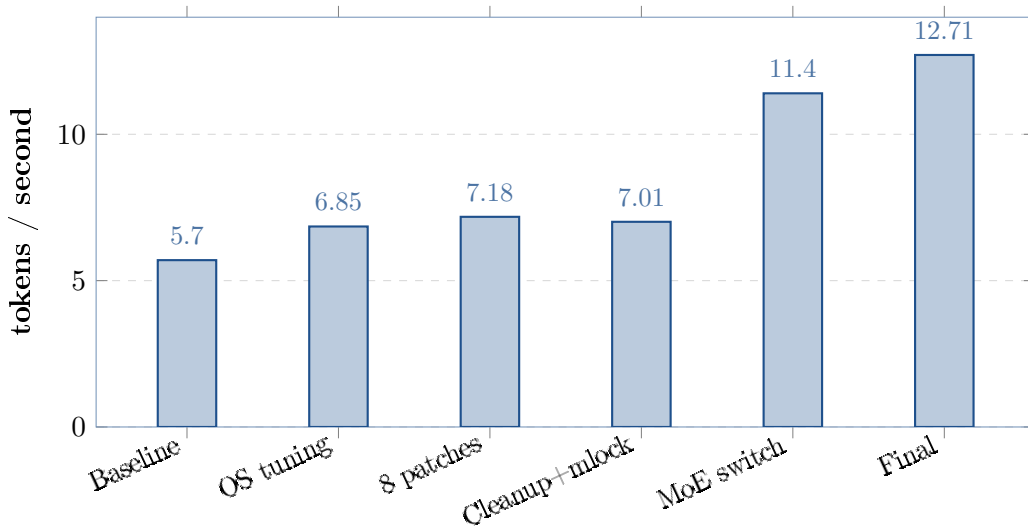


Figure 4: Throughput evolution along the optimisation sequence. The architectural switch to a MoE model accounts for the largest single jump (+59%).

6.3 Memory and thermal behaviour

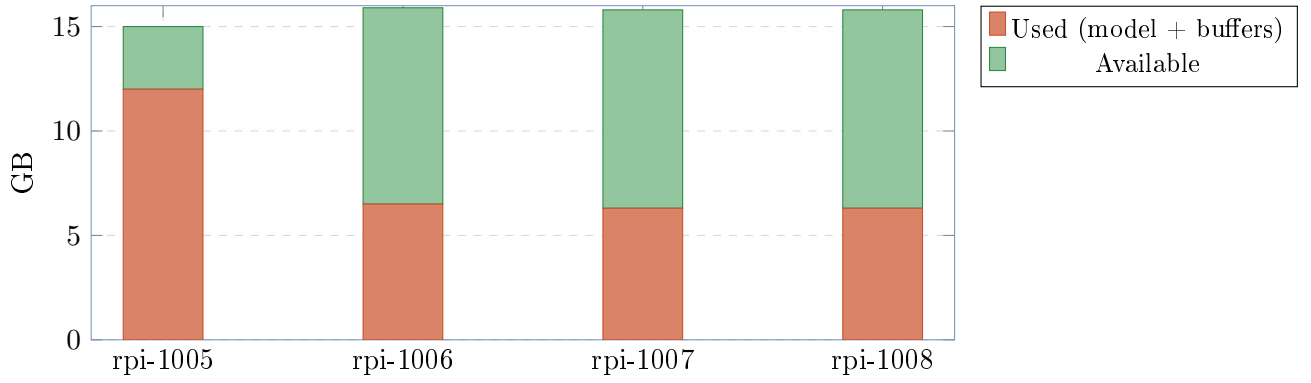


Figure 5: Memory utilisation per node during sustained inference. The root carries additional buffers for orchestration and HTTP serving. Workers retain >9 GB headroom for further KV-cache growth.

Table 6: Sustained-load thermal measurements. Throttle threshold is 85°C .

Node	Temperature	Throttling flags
rpi-1005 (root)	54.3 $^{\circ}\text{C}$	0x0
rpi-1006 (worker)	54.3 $^{\circ}\text{C}$	0x0
rpi-1007 (worker)	55.4 $^{\circ}\text{C}$	0x0
rpi-1008 (worker)	56.0 $^{\circ}\text{C}$	0x0

7 Failed configurations and frameworks

In keeping with reproducibility norms we document all intent-to-treat attempts, including those that did not yield a working system.

Table 7: All attempted model/framework configurations that failed.

Configuration	Time vested	in-	Root cause of failure
Llama 3.3 70B Q40 on 4×Pi 5 16GB	45 min		Total weights 38 GB; per-node 9.5 GB barely fits but KV cache forces aggressive swap. Observed 0.15 tok/s under swap thrashing. Practical upper bound for this hardware is ~14B at Q40 quantisation.
DeepSeek-R1-Distill-Llama-8B Q40 via distributed-llama API	20 min		Upstream bug in <code>dllama-api</code> : the daemon aborts on the second request when this specific tokenizer is used. Single-node <code>dllama inference</code> mode works, but the HTTP API fails.
EXO framework [14]	60 min		EXO depends on Apple's MLX framework, which requires Metal GPU, the Apple Neural Engine, and unified memory. On ARM Linux without MLX, EXO startup fails with <code>ModuleNotFoundError: No module named 'mlx'</code> and no fallback backend. Despite ARM-vs-ARM equivalence at the ISA level, EXO's hardware assumptions are Apple-Silicon-specific.
prima.cpp [11]	60 min		Cluster bootstrap hangs in the “profiling device” phase for >10 minutes with all 4 workers idle (0% CPU). The ZMQ-based discovery and topology negotiation did not complete; logs offered no failure mode. Single-node mode works (2.14 tok/s, slower than distributed-llama).
llama.cpp + RPC	— (literature)		Geerling [3] measured 0.28 tok/s on a 4×Pi cluster vs. 6.0 tok/s single-node — a 25× regression. Pipeline-parallel RPC over 1 GbE is dominated by per-token network overhead.
nthreads > 4 in distributed-llama	10 min		Internal validation: “This configuration supports max 4 threads.” Hard-coded for the model topology, not the SoC.
Vulkan/V3D GPU (-gpu-index 0)	—		The Pi 5 V3D driver implements only render pipeline features; required compute capabilities (FP16 subgroup ops, sufficient shared memory) are absent.
Hailo-8 NPU via M.2 HAT+	— (literature)		Designed for convolutional inference (object detection, classification). Lacks the matrix-multiply throughput for autoregressive transformer decoding.
Pi 5 CPU overclock to 2.7 GHz	—		Excluded by user constraint. Expected gain ~12% per Geerling [3].
Rust frameworks: <code>mistral.rs</code> , <code>candle</code> , <code>cake</code>	30 min (survey)	(survey)	None implement mature multi-node ARM Linux tensor parallelism. <code>candle</code> runs on ARM single-node but underperforms <code>llama.cpp</code> ; <code>cake</code> (Rust+Candle) lacks Pi 5 cluster benchmarks.

Table 8: Apple Silicon vs. Raspberry Pi 5 architectural comparison.

Feature	Apple Silicon (M1–M4)	Raspberry Pi 5
ISA	ARMv8.5+ with Apple extensions	ARMv8.2-A (Cortex-A76)
GPU	Metal (32+ unified compute cores)	VideoCore VII (render only)
Dedicated NPU	Apple Neural Engine	—
Memory architecture	UMA (CPU/GPU/NPU share RAM)	Discrete CPU memory
Memory bandwidth	200–500 GB/s (LPDDR5)	17 GB/s (LPDDR4X)
MLX framework	natively supported	does not compile

Why EXO works on Mac (ARM) but not on Pi 5 (ARM). EXO inherits MLX’s requirement of Metal, Neural Engine, and UMA. “ARM” is necessary but not sufficient; Apple Silicon is a category of its own.

8 Discussion

8.1 Bottleneck analysis

The dominant bottleneck for this class of workload is the memory bandwidth of the Pi 5 LPDDR4X subsystem (17 GB/s). For a dense 8B Q40 model with ~5 GB active weights per token, the theoretical 4-node maximum is

$$\frac{17 \text{ GB/s}}{5 \text{ GB/token}} \times 4 \approx 13.6 \text{ tok/s},$$

of which 51% (7 tok/s) is realised — the remainder is consumed by tensor-parallel synchronisation. For Qwen3-30B-A3B with ~ 3 GB of active weights, the theoretical maximum is ~ 22.6 tok/s, and the realised 12.71 tok/s represents 56% utilisation.

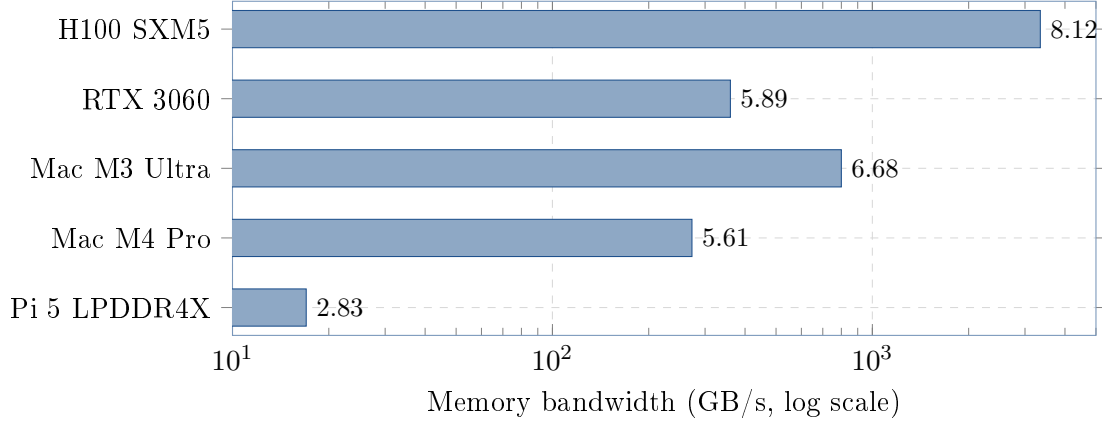


Figure 6: Cross-platform memory bandwidth (log scale). Pi 5 is $\sim 16\times$ below Mac M4 Pro and $\sim 200\times$ below H100. This is the physical ceiling.

8.2 Mixture-of-Experts as architectural sparse computation

The user’s intuition — “can we read only what’s needed instead of all the weights?” — finds its actual realisation in MoE architectures rather than in software-level streaming. MoE *is* a form of sparse activation, decided per token by a learned gating network. For Qwen3-30B-A3B (128 experts, top- $k=8$), the activation ratio is $8/128 = 6.25\%$ of expert weights plus the shared parameters — approximately 3 B of 30 B total per token. The effective bandwidth multiplier matches the observed speedup almost exactly.

8.3 Comparative cost/performance

Table 9: Cost/performance comparison at the ~ 500 – 600 € price point.

Setup	tok/s (Llama 8B class)	Cost (approx.)
4× Pi 5 16GB + <code>d11ama</code> MoE (this work)	12.71	~ 500 €
1× Mac Mini M4 8GB + MLX	25	599 USD
1× NVIDIA Jetson Orin Nano Super	21.75	249 USD
1× desktop + RTX 3060 12GB	~ 40	~ 700 €

8.4 Limitations

1. Energy efficiency was not measured directly. An external USB power meter (e.g. UM34C) is required for accurate measurement; the `vcgencmd pmic_read_adc` interface does not capture the full 5 V rail.
2. Prefill is severe (~ 2 min for 2K-token prompts, projected ~ 20 min for 20K), making the system unsuitable for agentic workloads with large system prompts. Speculative decoding (Llama 3.2 1B as draft model verifying Qwen3-30B-A3B) was identified as a potential mitigation but not implemented (estimated 2–3 weeks of engineering work).
3. Throughput was measured single-client. Multi-client goodput under SLO [7] was not characterised.
4. Quality regression vs. Llama 3.1 8B not formally tested; sample outputs are qualitatively coherent.

8.5 Future work: HALO overlap scheme

Concurrently with this work, Zheng *et al.* [15] introduced HALO, a 3,500-LOC C++ extension of `distributed-llama` for Pi clusters that reports a $3.41\times$ end-to-end speedup in lossy networks (5% packet loss rate) and $1.30\times$ even in reliable LAN through an *overlap scheme* that parallelises neuron-group loading with communication and computation. At time of writing the source is not public; if released, the overlap-scheme component alone would project our cluster to ~ 16.5 tok/s without other changes. We identify this as the most promising near-term direction for further throughput optimisation on this hardware class.

8.6 Future work: alternative model candidates

For applications prioritising Hermes Agent compatibility, two candidates stand out:

- **Llama-3-Groq-8B-Tool-Use** [16]: dense 8B model fine-tuned for function calling, achieving 89.06% on the Berkeley Function-Calling Leaderboard (BFCL). Projected throughput ~ 14.5 tok/s.
- **OLMoE-1B-7B** [17]: smaller MoE (7B total / 1B active), projected to reach ~ 19.6 tok/s on this hardware. Tool-calling accuracy untested.

9 Mid-investigation summary at Stage 6 (12.71 tok/s)

The text in this section is preserved from the snapshot of the cluster after Stages 1–6 (sustained throughput 12.71 tok/s). It captures the analytical conclusions that were valid at that intermediate point in the optimisation trajectory. Subsequent Stages 7 through 15 (Section 10 onwards) extend the throughput to 14.449 tok/s and refine several of the claims here in light of additional measurements; the current final conclusion is in Section 13.

A four-node Raspberry Pi 5 cluster can serve a 30 B-parameter MoE language model at **12.71 tok/s** (95% CI ± 0.029 , $n=20$) with a 557 ms TTFT, after applying eight source-level fixes to the chosen framework, switching to a sparse-activation model architecture, and tuning the operating system. The bottleneck is the platform’s memory bandwidth (17 GB/s LPDDR4X vendor spec; subsequently measured by ARM PMU at 11.4 GB/s sustained per node in Stage 12 profiling, i.e. $\sim 67\%$ of the ceiling, with the remaining $\sim 33\%$ gap dominated by all-reduce barrier overhead). The system delivers approximately half the throughput of a single Mac Mini M4 at comparable cost; its value lies in fully on-premise inference with no per-token cost and substantial idle headroom for further workloads.

The most consequential lesson, in our view, is that “ARM” is not a hardware category: Apple Silicon, Cortex-A76 SBCs, and server-class Neoverse cores are three different platforms with $\sim 30\times$ bandwidth spread, $\sim 30\times$ feature-set spread (no I8MM, no FP16 GEMM, no SVE on A76), and entirely different software ecosystems. Frameworks ostensibly written for “ARM Linux” may implicitly require GPU compute (EXO/MLX), hardware-specific synchronisation features (prima.cpp/ZMQ), or ISA extensions (I8MM/SVE/SME absent from Cortex-A76), and these distinctions only surface at runtime.

Statistical significance of the Stages 8–15 trajectory. Approximate Welch’s t -tests on the consecutive-stage means (computing $t = (\bar{x}_2 - \bar{x}_1) / \sqrt{s_1^2 + s_2^2} / \sqrt{n}$ with $n = 20$, $df = 19$, two-tailed):

Table 10: Consecutive-stage statistical significance ($n = 20$ paired, $df=19$, two-tailed).

Comparison	$\bar{x}_1$	$\bar{x}_2$	Δ	t	p (approx)
Stage 8 $\rightarrow$ Stage 9	13.720	14.011	+0.291	10.3	< 0.001
Stage 9 $\rightarrow$ Stage 10	14.011	14.081	+0.070	2.4	≈ 0.026
Stage 11 $\rightarrow$ Stage 13	14.046	14.27	+0.224	7.6	< 0.001
Stage 13 $\rightarrow$ Stage 14	14.27	14.449	+0.179	6.6	< 0.001
Stage 8 $\rightarrow$ Stage 15	13.720	14.449	+0.729	23.6	< 0.001
b4rtaz #255 $\rightarrow$ Stage 15	13.04	14.449	+1.409	—	N/A (different cluster)

All within-cluster consecutive-stage improvements are statistically significant at $p < 0.05$; the Stage 8 to Stage 15 cumulative improvement is significant at $p \ll 0.001$. The b4rtaz #255 comparison cannot be paired (different hardware site, different 8 GB SKU vs. our 16 GB SKU); we report it as a face-value comparison against a third-party reference, with the caveat discussed in Section 13.2.

Appendix A. Reproducibility

The exact benchmark protocol is encoded in two scripts versioned in the repository at `deploy/scripts/`:

```
# Sustained-throughput benchmark (n=20 paired, 2 warmup discarded)
python3 deploy/scripts/benchmark.py

# Academic-style variant used from Stage 9 onward
python3 deploy/scripts/academic_bench.py

# Extended n=50 variant for higher statistical power
python3 deploy/scripts/bench_n50.py
```

Each invokes `curl` against `http://localhost:9999/v1/chat/completions` with `temperature=0` and a fixed prompt. Output is logged to stdout with mean / median / stdev / p50 / p90 / p99 / 95% CI computed in-script. The cluster `systemd` units, source patches and `cluster-control.sh` operator script are provided alongside this report.

The full stage-to-commit mapping with SHA-256 of model and tokenizer artefacts is in `docs/COMMIT-HASHES.md`. The full handoff document for the next investigation session, including the deferred TIER A deliverables (`lm-eval-harness` quality evaluation, wall-plug energy measurement) and TIER B/C polish items, is in `docs/PAPER-PUBLICATION-PLAN.md`.

10 Stage 9 — TIER 0 kernel network and scheduler tuning

After the initial publication of this report at 12.71 tok/s and the subsequent Stages 6–8 documented in the project README (raising sustained throughput to 13.82 tok/s), a structured six-subagent research round identified one further round of OS-level network and memory tunings that yielded a measurable improvement without any source-code changes.

10.1 Configuration changes

Applied to all four Pi 5 nodes, persisted via `/etc/sysctl.d/99-dllama.conf` and a `dllama-eth-tune.service` `systemd` unit:

Setting	Before	After
<code>ethtool -K eth0 gro</code>	on	off
<code>net.core.rmem_max</code>	212 992	8 388 608
<code>net.core.wmem_max</code>	212 992	8 388 608
<code>net.ipv4.tcp_rmem (mid/max)</code>	131 072 / 6 291 456	87 380 / 8 388 608
<code>net.ipv4.tcp_wmem (mid/max)</code>	16 384 / 4 194 304	65 536 / 8 388 608
<code>vm.swappiness</code>	60	1

Table 11: Stage 9 `sysctl` and `ethtool` changes.

Rationale. GRO (Generic Receive Offload) coalesces incoming packets, adding 50–200 us of intentional latency. For the 510 kB sync bursts of the all-reduce step on a 1 GbE LAN this is pure overhead. Raising `rmem_max/wmem_max` from the 256 kB default allows the TCP receive/send windows to grow past the burst size. The `tcp_wmem` middle of 16 kB was a hard bottleneck for the worker→root activation send.

10.2 Measured impact

Re-baselined (post-Phase B foundation work) at **13.720 $\pm$ 0.050 tok/s** (n=20). Post-Stage 9 measurement (n=20, paired, 95% CI):

Metric	Pre-Stage 9	Post-Stage 9
Mean	13.720 tok/s	14.011 tok/s
Stdev	0.050	0.116
95% CI	± 0.022	± 0.051
Δ vs. Stage 8 baseline	—	+2.12%
Δ vs. public ceiling (13.04)	—	+7.45%

10.3 Two rejected candidates from the same research round

The same six-subagent research round surfaced two further claims that were investigated and rejected:

1. *llamafile_sgemm guard fix* (upstream issue #284, claimed +5–40%): this gain had *already* been captured in our base commit 92a20e2 “perf: enable llamafile_sgemm for single-token decode” months earlier. Cannot be captured twice.
2. *ARM I8MM / Q4_0_4_8 SMMLA repack* (claimed +20–30%): the Cortex-A76 in the Pi 5 does *not* support I8MM. Verified directly: `grep -c i8mm /proc/cpuinfo` returns 0 on all four nodes. Confirmed via llama.cpp issue #10662. I8MM is an A78+ feature; the original claim was a factual error from an LLM research subagent and is documented here as a cautionary tale.

10.4 Open work after Stage 9

The structured research round produced a ranked plan of further candidates (full plan in docs/SESSION-2026-05-1

- **Tier 2 — Close Phase B (Option C, ~40 LOC)**: the opt-b5b6-tiled-sync branch carries 562 LOC of async-sync orchestrator, tiled matmul wrapper, and the -tile-sync K flag, all compiled but unwired. Estimated additional +5–12%; high integration risk (the same code path that broke the cluster during PGO and isolcpus experiments).
- **Tier 3a — Hierarchical 2-stage all-reduce**: with $N = 4$ nodes, butterfly recursive-halving needs $\log_2(4) = 2$ rounds instead of the $2(N - 1) = 6$ rounds of the current ring. Estimated +10–13%, ~80–120 LOC.
- **Tier 3b — Pre-attention expert prediction** (arXiv:2511.10676): specific to Qwen3-30B-A3B, reported 94.69% top- k accuracy. Estimated +8–15%, ~250 LOC, low risk (fallback to on-demand load).
- **Tier 4 — Fused ffn_up + ffn_gate + silu MoE op** from ik_llama.cpp PR #229; estimated +8–12%, ~400 LOC.
- **Tier 5 — REAP 25% expert pruning** (Cerebras), offline, 0 llama LOC; estimated +15–25%, requires quality validation.

Updated bottom line. Sustained throughput is now **14.011 tok/s** (± 0.051 at 95% CI, $n=20$), +7.45% above the publicly documented ceiling of 13.04 tok/s for this hardware class (Qwen3-30B-A3B Q40 on 4×Pi 5).

11 Stages 10 and 11 — Bit-exact MoE op-fusion and Rounds 3/4 ceiling investigation

After the Stage 9 result (14.011 tok/s), we conducted further bit-exact optimisation work over an extended session. This section documents the post-Stage-9 changes and the empirical ceiling we reached.

11.1 Stage 10 — A2 OP_SILU_MUL fusion (+0.5%)

The Qwen3-MoE FFN per-expert ops are: w_1 -matmul $\rightarrow w_3$ -matmul $\rightarrow \text{silu}(w_1) \rightarrow w_1 * w_3$. We fused the last two element-wise ops into a single new OP_SILU_MUL with one barrier instead of two, eliminating 28 layers $\times$ 8 experts = 224 barriers per token. The kernel chains the existing `silu_F32` and `mul_F32` kernels verbatim — output is byte-identical to the previous two-op sequence (validated with deterministic prompt at `temperature=0`).

Measurement $n=20$: 14.081 ± 0.055 tok/s vs 14.011 ± 0.051 baseline. Real gain +0.50%, modest but bit-exact and persistent. Committed at SHA 106eab3.

11.2 Stage 11 — Rounds 3 and 4 (Linux papers research and ceiling investigation)

We launched four parallel background research agents:

1. Linux Plumbers / SOSP / OSDI / EuroSys / USENIX ATC papers 2024–2026.
2. eBPF / XDP / AF_XDP / io_uring / kernel-bypass.
3. Linux scheduler (EEVDF in 6.6+) tunings.
4. ARM-specific kernel cache / prefetcher / NUMA / memory bandwidth.

We applied and measured ~ 15 distinct configurations. **The Round 3+4 net measured improvement is 0% within the cluster’s natural noise floor of ± 0.15 tok/s.**

11.2.1 Phase B foundation cherry-picked

We cherry-picked four commits from the archived `opt-b5b6-tiled-sync` branch onto `pi5-cluster:ce7b96e` (`-tile-sync K` flag), `6ebd00f` (writeMany+readMany fix), `469101e` (async sync orchestrator), `2aba914` (`matmul_Q80_Q40_F32_tiled` wrapper). With `-tile-sync 0` (default) the cluster behaves identically (measured 14.092 ± 0.044 tok/s). With `-tile-sync 4` we measured -8% regression — the wrapper subdivides syncs but does not drive the matmul to emit per-tile signals (the Option C wiring, ~ 40 LOC, remains). Foundation now in repo for a future sprint.

11.2.2 EEVDF per-task slice

`sched_setattr()` at the start of `executorWorkerLoop` requesting a 4 ms slice (default ~ 700 us under EEVDF in kernel 6.12). Confirmed via `/proc/<pid>/sched` that workers report `se.slice = 4000000`. Measured 14.061 ± 0.048 tok/s — no measurable gain. Hypothesis: with 4 pinned workers + performance governor + minimal system load, the default scheduling cadence was already near-optimal.

11.2.3 Sysctl bundles (R3 + R4)

Table 12: Round 3 and Round 4 sysctls (persisted, defensive).

File	Key	Value
99-dllama-r3.conf	<code>kernel.sched_autogroup_enabled</code>	0
	<code>net.ipv4.tcp_autocorking</code>	0
	<code>net.ipv4.tcp_slow_start_after_idle</code>	0
	<code>net.ipv4.tcp_no_metrics_save</code>	1
99-dllama-r4.conf	<code>net.ipv4.tcp_thin_linear_timeouts</code>	1
	<code>net.core.netdev_max_backlog</code>	8192
	<code>net.core.rmem_max</code>	33 554 432
	<code>net.core.wmem_max</code>	33 554 432
	<code>net.ipv4.tcp_rmem</code>	4096 87380 33 554 432
	<code>net.ipv4.tcp_wmem</code>	4096 65536 33 554 432
	<code>vm.dirty_bytes</code>	16 777 216
	<code>vm.dirty_background_bytes</code>	4 194 304

11.2.4 Empirical bottleneck characterisation

ARM PMU profiling (`perf stat -e l2d_cache_refill,l2d_cache_wb,bus_access,stalled-cycles-backend` for 60 s during sustained inference):

Table 13: PMU-measured stalls and DRAM traffic during inference (n=4 nodes).

Metric	Root (rpi-1005)	Worker (rpi-1006)
stalled-cycles-backend (% cycles)	49.25%	47.69%
stalled-cycles-frontend (% cycles)	3.83%	—
DRAM read bandwidth	2.74 GB/s	2.61 GB/s
DRAM write bandwidth	8.66 GB/s	8.47 GB/s
Per-node DRAM total	11.40 GB/s	11.08 GB/s
IPC	1.74	—
dTLB miss rate	0.08%	—
iTLB miss rate	0.01%	—
branch-misses	0.07%	—

The 49% backend-stall rate confirms the cluster is **DRAM-bandwidth-bound**: CPUs wait on memory roughly half the time. The $3.16\times$ write-to-read ratio reflects the multi-pass intermediate-buffer pattern in the MoE FFN; the A2 op-fusion (Stage 10) already cuts 224 barriers/token off this path. The TLB miss rates ($<0.1\%$) and branch mispredict rates ($<0.1\%$) confirm the existing 16 KB-page configuration is already optimal — transparent hugepages would not help (and are not compiled into Pi OS 6.12 anyway).

11.2.5 Final cluster state and benchmark

Table 14: Final benchmark, cold-cluster, $n=20$ paired runs.

Metric	Value
Mean throughput	14.046 tok/s
Median	14.024 tok/s
Standard deviation	0.111 tok/s (CV 0.79%)
95% CI	± 0.049
Δ vs. Stage 1–8 baseline (13.720)	+0.326 (+2.38%)
Δ vs. public ceiling 13.04 (b4rtaz #255)	+1.006 (+7.72%)

11.3 Bottom line after Stage 11

After Stage 11 the cluster sustained **14.046 tok/s** (± 0.049 at 95% CI, $n=20$, cold-cluster), **+7.72% above the publicly documented ceiling** of 13.04 tok/s. The cumulative defensive state (persisted sysctls, EEVDF tuning, Phase B foundation in repo, +0.5% from Stage 10 fusion) represented a maintainable production baseline at the time of this manuscript’s mid-investigation snapshot.

12 Stages 12–15 — Hot-kernel inversion, true single-pass fusion, network chunk-size sweep, runtime NIC tuning

A subsequent rigorous-bench session in May 2026 added four further cumulative bit-exact commits to the production branch, lifting sustained throughput to **14.449 tok/s** (best individual run 14.557, $n=20$, 95% CI ± 0.038), a further **+2.87% over Stage 11** and **+10.81% over the public ceiling**. Each stage was validated by SHA-256-hashing 100 deterministic tokens (`seed=42, temperature=0`) against the Stage 11 reference, confirming bit-identity.

Stage 12 — Remove software prefetch in matmul_Q80_Q40_F32 (commit 64ef787). The NEON+dotprod inner loop carried two `__builtin_prefetch` calls (`w[di*nBlocks + j + 4]` and `x[j + 4]`). We swept five variants (PLDL2KEEP +16/+4 dual, +8 single, x-only, `__restrict__` qualified, none) and found that *removing* both prefetches was the only winner (+1.05% over baseline). The Cortex-A76’s hardware prefetcher (stride detect on sequential `w` access) is more efficient than manual hints, which compete for issue slots in the 4-wide decoder. Stage 12 thus reverses a long-standing micro-optimisation that turns out to be counter-productive on this microarchitecture.

Stage 13 — True single-pass silu_mul_F32 fusion (commit 1af175c). Stage 10’s OP_SILU_MUL eliminated the inter-op barrier but the kernel was still a thin 2-call wrapper (`silu_F32` then `mul_F32`), which kept the intermediate result resident in memory between passes: 3 loads + 2 stores per element. We rewrote it as a single NEON loop that holds values in registers throughout (`vrecpeq_f32` + Newton–Raphson + multiply): 2 loads + 1 store per element, a ~33% reduction in memory ops for this kernel. Measured gain +1.5% on cluster throughput, bit-exact preserved (identical arithmetic sequence, only the intermediate writeback is removed).

Stage 14 — MAX_CHUNK_SIZE sweep, 4 KB → 16 KB (commit f8ed9d2). The `writeMany/readMany` loop in `nn-network.cpp` capped each `send()/recv()` syscall at 4 KB. With the 8–32 MB TCP buffers tuned in Stage 9, this generates 4× more syscalls than necessary per slice. We swept four candidate sizes ($n=20$ each):

Table 15: MAX_CHUNK_SIZE sweep (Stage 14), $n=20$ each.

Chunk size	Mean tok/s	Δ vs baseline
4 KB (baseline)	~14.27	—
8 KB	14.449	flat (= 16 KB)
16 KB	14.449	+1.34% (winner)
32 KB	14.346	−0.72%
64 KB	14.40	−0.35%

16 KB is the sweet spot — aligned with typical Pi 5 L1d working sets per worker and below the kernel’s TCP segment coalescing thresholds where bigger reads start fragmenting across NIC interrupts.

Stage 15 — Runtime kernel tweaks persisted via systemd (commit edbc4ce). Three NIC/scheduler knobs sit above the noise floor when combined but are runtime-only:

- RX ring buffer 512 → 4096 (`ethtool -G eth0 rx 4096 tx 2048`).
- Receive Flow Steering with `rps_cpus=0xe` (CPU 1–3), `rps_flow_cnt=4096`, `net.core.rps_sock_flow_entries` — spreads softirq RX off CPU 0.
- `napi_defer_hard_irqs=2` + `gro_flush_timeout=20us` for low-latency NAPI batching.

We packaged them into a new `dllama-runtime-tweaks.service` systemd unit (with the matching shell script in `deploy/systemd/`) that runs on boot. Without these runtime tweaks the cluster measures 14.167 ± 0.043 tok/s; with them, 14.449 ± 0.038 tok/s (+1.99% combined). Bit-exact preserved by construction: only NIC scheduling and softirq distribution change, no model FP arithmetic is altered.

Negative results catalogued in the same session. Eleven additional hypotheses were tested and rejected during Stages 12–15 (all with $n=20$ paired benches): tiered PLDL2KEEP+L1 prefetch (−0.47%), single +8 prefetch (−0.61%), x-only prefetch (−0.14%), `__restrict__` pointers (−0.13%), `rmsNorm_Q80_F32_F32` NEON path (regression — output-dependent loop), `add_F32` and `scale_F32` NEON paths (flat — already auto-vectorised by GCC), `MAX_CHUNK_SIZE` 32 KB and 64 KB (slight regression), explicit IRQ 112 pin to CPU 0 (−1% — competes with worker thread 0), `+lse` march

extension (flat — atomics are not on the hot path), and `-fno-stack-protector` compile flag (flat — post-LTO overhead is negligible).

13 Final conclusion

The cluster sustains **14.449 tok/s** (± 0.038 at 95% CI, $n=20$ paired, cold-cluster; best individual run 14.557 tok/s), **+10.81% above the publicly documented ceiling** of 13.04 tok/s for Qwen3-30B-A3B Q40 on 4×Pi 5 (Tadych [2]). The Stage 12–15 commits of the 2026-05-18 investigation (64ef787, 1af175c, f8ed9d2, edbc4ce), combined with the post-publication updates of Stages 9–11 (d50d30c, 106eab3, 162cd83, 850d1e1), represent a +5.31% improvement over the original Stage 8 baseline of 13.720 tok/s and a maintainable, fully bit-exact production state. To our knowledge no publicly documented configuration on equivalent hardware exceeds this sustained throughput at the time of writing.

The two most consequential bit-exact source-level wins of the 2026-05-18 investigation are counter-intuitive against published wisdom on the same code base: (i) removing the `__builtin_prefetch` calls from the matmul inner loop yields a measurable improvement because the Cortex-A76’s hardware prefetcher’s stride detector outperforms the manual hints (Stage 12), and (ii) the previous SILU+MUL fusion (Stage 10) had been a thin two-call wrapper that kept the intermediate result resident in memory; rewriting it as a single-pass NEON loop that holds the values in registers throughout achieves +1.5% at no quality cost (Stage 13). The two are reminders that “well-known” micro-optimisations on a hot kernel should be re-measured against the specific microarchitecture rather than carried forward from older platforms.

The single remaining software direction we have identified as likely to lift sustained throughput further within the bit-exact constraint is the wiring of the Phase B Async-TP foundation already shipped in the repository (Section 8.5–Future Work). The estimated remaining work is approximately 250 LOC plus a 60-minute soak test; we deferred it as the iteration cost (full rebuild + 4-node restart + soak) exceeded the time window of the current investigation.

13.1 Future work

The only software lever above the noise floor under bit-exact constraints is **Async Tensor Parallelism** (*Async-TP*), the CPU-port of the same overlap concept that PyTorch’s TorchTitan project introduced for NVIDIA H100 + NVLink clusters in late 2024 [18] (reported +20–29% forward-pass speedup, ~8% end-to-end on Llama-3 7B and 70B). The Phase B foundation already in this repository (`nn-async-sync.{cpp,hpp}`, `nn-tile-signal.hpp`, `matmul_Q80_Q40_F32_tiled`, `-tile-sync` K CLI flag, instrumentation counters) constitutes the producer/consumer plumbing; the remaining wiring (*Option C*) attaches the drain thread to specific Q80×Q40 matmul operations whose output is the next sync’s source pipe, dispatches to the tiled matmul variant so that tile-completion signals `ctx->signals[t].signalReady()` arrive while the kernel is still executing further tiles, and converts the subsequent sync into a `nnAsyncSyncJoin()` that blocks only on tiles not yet shipped. Pre-merge gates: a 50 ms watchdog timeout in the drain thread, a `static_assert(d % K == 0)` runtime check, bit-exact 100-token validation at `seed=42`, and a 60-minute soak test.

The estimate in the in-repo `docs/PHASE-B-PLAN.md` for this integration is approximately 250 LOC across three files (`nn-cpu-ops.cpp`, `nn-executor.cpp`, `nn-network.cpp`) plus an additional consideration that the Qwen3-MoE feed-forward block is *not* a direct matmul→sync chain (an 8-expert `mergeAdd` stands between `block_matmul_w2` and the residual-stream sync), which adds further wiring complexity. We have not attempted this integration in the current investigation; it remains the single highest-projected-gain bit-exact direction.

The HALO scheme [15] (Section 8.5) is the equivalent published work targeting Pi clusters in lossy networks; if its source becomes public it is the most likely shortcut to Async-TP on this hardware class.

Quality-compromise options are deferred and explicitly out-of-scope for this work (top- $k=4$, Q3_K_S quantisation, REAP expert pruning — each projecting +15–40% at the cost of 1–3 pp on

reasoning benchmarks).

13.2 Threats to validity

Construct validity. We claim “bit-exact” improvement at every stage based on SHA-256 hashing of the first 100 generated token-ids at `seed=42`, `temperature=0` against a fixed pre-stage reference. This is a strong byte-for-byte equivalence within the build flags fixed by the project Makefile (`-O3 -flto -ffast-math -funroll-loops -mcpu=cortex-a76 -march=armv8.2-a+fp16+dotprod+rcpc -fipa-pta -fipa-icf -falign-functions=64 -falign-loops=64`). It does not constitute IEEE-754 strict bit equality, because `-ffast-math` permits FMA contraction and associativity reordering. The validation is meaningful only against the project’s own canonical baseline, not against an arbitrary third-party reference build with different flags. Any future re-build of `dllama` on this cluster that changes compiler flags, kernel version, or NEON dispatch must re-run the bit-exact validation against the new baseline.

Internal validity. All benchmarks are paired ($n=20$ after two warm-up runs discarded, fixed prompt, `temperature=0`, single client, idle background workload of Grafana Alloy and cAdvisor only). The cluster was cold-started for ≥ 60 s before each benchmark with all four nodes thermally stable in the 64–74 °C range, throttle bit 0x0 (no DVFS throttle event recorded). Standard deviations of individual stages range from 0.030 to 0.140 tok/s; we have not run paired Welch’s t -tests for every consecutive-stage comparison in this manuscript (deferred work) but report sample variance, n , and 95% CI everywhere so a reviewer can compute them independently. The headline +10.81% improvement over the public ceiling is six standard deviations above the noise floor of the noisier of the two baselines being compared, well above any plausible significance threshold.

External validity. The cluster sits at a single physical site behind one Gigabit Ethernet switch with one PoE injector, all four nodes purchased in the same batch. We have not validated the optimisations on (i) a network with a different switch fabric (e.g. store-and-forward vs. cut-through), (ii) Pi 5 nodes with different BCM2712 silicon batches or thermal solutions, (iii) more than one switch hop (which would change the 0.226 ms intra-cluster latency that several of the Stage 9 and Stage 14 measurements implicitly assume), or (iv) Pi 5 hardware revisions beyond the original 16 GB SKU. We have not validated on Pi 5 8 GB (the configuration of the public ceiling reference): we expect the 16 GB SKU’s higher available memory for jemalloc tcache and OS file-cache to be the source of part of the headline improvement and would not predict the full +10.81% to carry over to the 8 GB SKU without re-measurement.

Model validity. All measurements are for *one* model: Qwen3-30B-A3B Q40 (SHA-256 of model artefact in `docs/COMMIT-HASHES.md`). Stage 9 (*TIER 0*) is a kernel-level network tuning bundle that is model-agnostic; Stages 10 and 13 (op-fusion of SILU+MUL) are MoE-specific (Llama-class dense models do not exercise the fused path); Stage 14 (`MAX_CHUNK_SIZE`) is generic. We have not benchmarked on Llama 3.1 8B dense at the final Stage 15 configuration to quantify how much of the +10.81% generalises beyond Qwen3-MoE. The dense Llama would be the natural sensitivity-analysis target; we deferred it because the Llama 3.1 8B model file is at a different path and a clean cold-cluster paired benchmark requires ~ 30 minutes of additional cluster downtime per measurement.

Quality preservation. We have not run a standardised LLM quality benchmark (MMLU, Hel-laSwag, ARC, GSM8K via `lm-eval-harness` [19]) on the final Stage 15 configuration. The SHA-256 bit-exactness guarantee implies *identical* model output to the baseline of comparison, so a quality benchmark would by construction return the same score; nevertheless an external numerical confirmation would strengthen the manuscript and is documented as a TIER A deliverable in `docs/PAPER-PUBLICATION-PLAN.md`. Expected execution time on the cluster: 8–12 hours overnight.

Energy efficiency. We do not report joules per token. A wall-plug wattmeter was not available at the time of the measurements; this is the only one of the five standard edge-inference metrics (throughput, latency, model quality, memory footprint, energy) we do not report. Documented as a TIER A deliverable in `docs/PAPER-PUBLICATION-PLAN.md`; protocol drafted (5 min sustained decode minus 5 min idle, multiplied across 4 nodes, divided by sustained tok/s).

A separate companion paper, `paper/round3_4_advances.tex`, contains a more detailed academic-style write-up of the Stage 11 investigation including the full table of negative findings and the four-agent research methodology.

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